

OPERATING SYSTEMS LAB

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS12	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To provide an understanding of the design aspects of operating system concepts through simulation
2. Introduce basic Unix commands, system call interface for process management, Inter-process communication and I/O in Unix

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Simulate and implement operating system concepts such as scheduling, deadlock management, file management and memory management.
2. Able to implement C programs using Unix system calls

LIST OF EXPERIMENTS

WEEK - 1	
	Write C programs to simulate the following CPU Scheduling algorithms a) FCFS b) SJF
WEEK - 2	
	Write C programs to simulate the following CPU Scheduling algorithms a) Round Robin b) priority
WEEK - 3	
	Write programs using the I/O system calls of UNIX/LINUX operating system (open, read, write, close, fcntl, seek, stat, opendir, readdir)
WEEK - 4	
	Write a C program to simulate Bankers Algorithm for Deadlock Avoidance and Prevention
WEEK - 5	
	Write a C program to implement the Producer – Consumer problem using semaphores using UNIX/LINUX system calls.
WEEK - 6	
	Write C programs to illustrate the following IPC mechanisms a) Pipes b) FIFOs
WEEK - 7	
	Write C programs to illustrate the following IPC mechanisms a) Message Queues b) Shared Memory

WEEK - 8	
Write C programs to simulate the following memory management techniques a) Paging b) Segmentation	
WEEK - 9	
Write C programs to simulate Page replacement policies a) FCFS	
WEEK - 10	
Write C programs to simulate Page replacement policies a) LRU	
WEEK - 11	
Write C programs to simulate Page replacement policies a) Optimal	
TEXT BOOKS	
- Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley - Advanced programming in the Unix environment, W.R.Stevens, Pears	
REFERENCE BOOKS	
- Operating Systems – Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI - Operating System - A Design Approach-Crowley, TMH. - Modern Operating Systems, Andrew S Tanenbaum, 2nd edition, Pearson/PHI - UNIX Programming Environment, Kernighan and Pike, PHI/Pearson Education - UNIX Internals: The New Frontiers, U. Vahalia, Pearson Education	